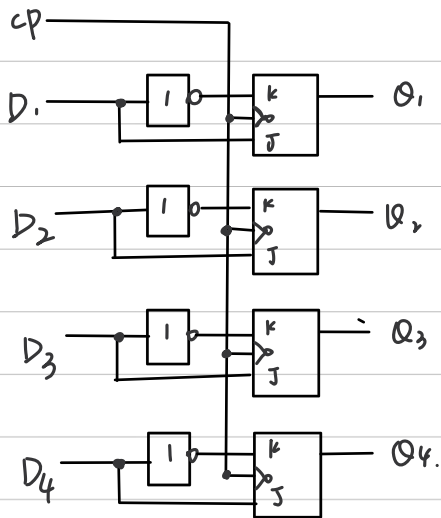


4.6.5

JK触发器的真值表.

J	K	Q_{n+1}
0	0	Q_n (保持.)
0	1	0
1	0	1
1	1	$\bar{Q}_n$

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n.$$



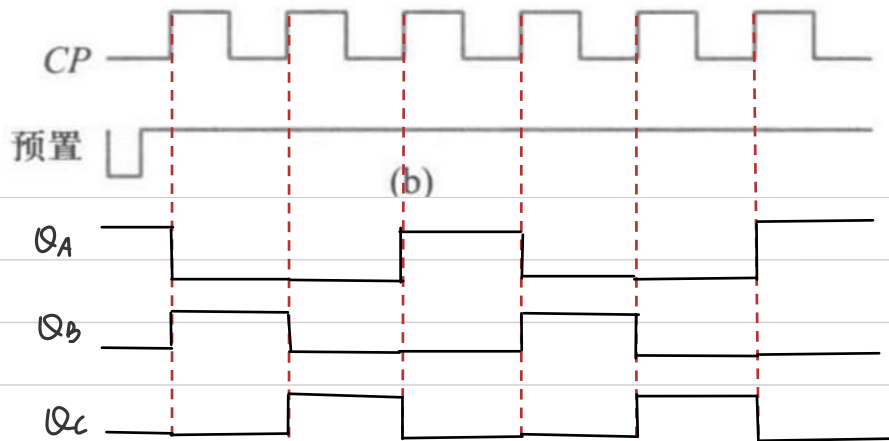
4.6.7

$$D_A = Q_C^n \quad D_B = Q_A^n \quad D_C = Q_B^n$$

$$Q_A^{n+1} = D_A = Q_C^n \quad Q_B^{n+1} = D_B = Q_A^n \quad Q_C^{n+1} = D_C = Q_B^n$$

初始状态:

$$Q_A Q_B Q_C = 100.$$



4.6.11

驱动方程: $J_0 = Q_2^n \oplus Q_3^n$ $K_0 = \overline{Q_2^n \oplus Q_3^n}$

$J_1 = Q_0^n$ $K_1 = \overline{Q_0^n}$

$J_2 = Q_1^n$ $K_2 = \overline{Q_1^n}$

$J_3 = Q_2^n$ $K_3 = \overline{Q_2^n}$

状态方程: $Q_0^{n+1} = Q_2^n \oplus Q_3^n$ $Q_1^{n+1} = Q_0^n$ $Q_2^{n+1} = Q_1^n$ $Q_3^{n+1} = Q_2^n$

状态转换表:

Q_0^n	Q_1^n	Q_2^n	Q_3^n	Q_0^{n+1}	Q_1^{n+1}	Q_2^{n+1}	Q_3^{n+1}
0	0	0	1	1	1	0	0
0	0	1	0	1	0	0	1
0	0	1	1	0	0	0	1
0	1	0	0	0	0	1	0
0	1	0	1	1	0	1	0
0	1	1	0	1	0	1	1
0	1	1	1	0	0	1	1
1	0	0	0	0	1	0	0
1	0	0	1	1	1	0	0
1	0	1	0	1	1	0	1
1	0	1	1	0	1	0	1
1	1	0	0	0	1	1	0
1	1	0	1	1	1	1	0
1	1	1	0	1	1	1	1
1	1	1	1	0	1	1	1
0	0	0	0	0	0	0	0

按 $Q_3 Q_2 Q_1 Q_0$ 顺序.

1000 $\rightarrow$ 0001 $\rightarrow$ 0010 $\rightarrow$ 0100 $\rightarrow$ 1001 $\rightarrow$ 0011 $\rightarrow$ 0110 $\rightarrow$ 1101
 $\swarrow$
 1100 $\leftarrow$ 1110 $\leftarrow$ 1111 $\leftarrow$ 0111 $\leftarrow$ 1011 $\leftarrow$ 0101 $\leftarrow$ 1010

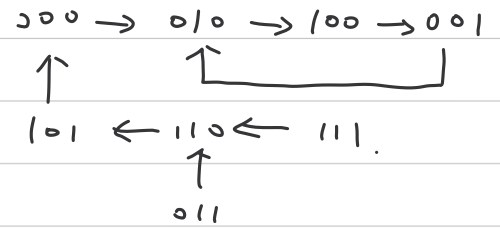
4.6.12.

驱动方程: $J_0 = Q_2^n$ $K_0 = 1$ $J_1 = \overline{Q_2^n}$ $K_1 = \overline{Q_0^n}$ $J_2 = Q_1^n$ $K_2 = \overline{Q_1^n}$

状态方程: $Q_0^{n+1} = Q_2^n \overline{Q_0^n}$ $Q_1^{n+1} = \overline{Q_2^n} \overline{Q_1^n} + Q_1^n Q_0^n$ $Q_2^{n+1} = Q_1^n$

表:

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}
0	0	0	0	1	0
0	1	0	1	0	0
1	0	0	0	0	1
0	0	1	0	1	0
0	1	1	1	1	0
1	1	0	1	0	1
1	0	1	0	0	0
1	1	1	1	1	0



4.6.13.

驱动方程 $J_0 = \overline{Q_1^n}$ $K_0 = Q_1^n$ $J_1 = Q_0^n$ $K_1 = Q_0^n$

状态方程 $Q_0^{n+1} = \overline{Q_1^n} \overline{Q_0^n} + Q_1^n Q_0^n = \overline{Q_1^n}$

$Q_1^{n+1} = Q_0^n \overline{Q_1^n} + Q_0^n Q_1^n = Q_0^n$

表:

Q_1^n	Q_0^n	Q_1^{n+1}	Q_0^{n+1}
0	0	0	1
0	1	1	1
1	1	1	0
1	0	0	0

00 → 11 → 10 → 00

$$\left\{ \begin{array}{l} \varphi_0 = \overline{cp} \overline{Q_1} \overline{Q_0} \\ \varphi_1 = \overline{cp} \overline{Q_1} Q_0 \\ \varphi_2 = \overline{cp} Q_1 Q_0 \\ \varphi_3 = \overline{cp} Q_1 \overline{Q_0} \end{array} \right.$$

